



STG-SurviNet: Mining Spatial, Temporal, and Survival Patterns in Urban Infrastructure Resolution Times Across Chicago Community Areas

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Abstract: Chicago's 311 system resolves 97% of infrastructure complaints, so whether repairs happen is largely settled. The more useful question is when and why some neighborhoods wait far longer than others. This study mines 644,554 complaint records from 2022 to 2026 across four infrastructure categories to surface three types of patterns embedded in that gap: spatial clusters of neighborhoods with structurally slow resolution times, temporal windows where backlogs spike, and survival risk profiles that vary by complaint type and area workload. To extract these patterns, the study develops STG-SurviNet, a hybrid architecture combining a Graph Convolutional Network (GCN) for spatial context, a Temporal Convolutional Network (TCN) for 60-day workload history, and a Deep Cox Proportional Hazards layer that produces individual resolution-time survival curves per complaint. The 19,547 open complaints are treated as right-censored observations rather than discarded. STG-SurviNet is used here as a structured lens, not as the research contribution itself, and the findings it surfaces about Chicago's infrastructure response patterns are the actual output of this study.

Keywords: Graph Convolutional Network, Survival Analysis, Cox Proportional Hazards, Chicago 311, Smart City.

Chapter 1 Introduction

Chicago, the third-largest city in the United States, operates one of the most well-documented public complaint systems in the world through its 311 Service Requests platform, which has accumulated over 13.9 million reports since its relaunch in December 2018 (City of Chicago, 2026). This study focuses on 647,521 infrastructure-related complaints filed between 2022 and 2026 across four categories: potholes, streetlight outages, tree debris, and traffic signal failures. Despite a resolution rate of 97%, the more operationally relevant question is not whether a complaint will be resolved, but when. The dataset reveals a heavily right-skewed distribution with a mean resolution time of roughly 22 days against a median of only 4 days, with some complaints remaining open for over 1,500 days. Unresolved infrastructure complaints carry real consequences: damaged roads pose

safety risks to motorists, non-functioning streetlights increase crime exposure at night, and prolonged response times erode public trust in city services. This gap also reflects identifiable spatial and temporal patterns, as community areas carrying high historical backlogs tend to resolve new complaints more slowly, and seasonal surges such as post-winter road damage can push departmental queues beyond capacity and compound delays across the entire system.

Conventional regression approaches cannot adequately model this problem for two reasons. First, the 19,547 complaints that were still open at the time of data extraction have no `closed_date`, meaning standard regression models must discard them entirely and introduce survivorship bias by learning only from already-resolved cases. Survival analysis addresses this directly by treating open complaints as right-censored observations, allowing them to contribute to

hazard estimation even without a known resolution time, a property that is central to deep survival frameworks such as DeepSurv (Katzman et al., 2018) and DeepHit (Lee et al., 2018). Second, regression treats each complaint independently, ignoring both the relational structure between neighbouring areas and the sequential load history that shapes how quickly a given area can respond. Graph Convolutional Networks have demonstrated strong capability in learning from graph-structured data where the state of each node is shaped by its neighbours (Defferrard et al., 2016; Kipf and Welling, 2017), making them well-suited for encoding the spatial pressure that overloaded areas exert on their surrounding community areas.

This study therefore proposes STG-SurviNet, a hybrid architecture that integrates a Graph Convolutional Network, a Temporal Convolutional Network, and a Deep Cox Proportional Hazards layer into a single end-to-end model trained with Cox Negative Partial Log-Likelihood loss. The study is guided by four research questions: first, how the spatial proximity structure between community areas influences resolution duration in neighbouring areas; second, whether the 60-day complaint backlog volume of an area correlates with changes in the resolution hazard of new complaints in that same area; third, whether the proposed GCN + TCN + Deep Cox architecture produces more accurate predictions than classical survival baselines including Kaplan-Meier, Cox PH without graph features, and Random Survival Forest; and fourth, which community areas consistently show low hazard scores and which spatial-temporal factors best explain those delays. To address these questions, this study aims to construct a spatial graph of Chicago's 77 community areas from official boundary data, build per-area daily complaint volume time series as sequential input to the TCN, train the full

architecture end-to-end in PyTorch, and evaluate it against five baselines using C-index, Integrated Brier Score, and D-Calibration. Academically, this work contributes a demonstration of integrating graph learning, sequence modelling, and survival analysis within one jointly trained architecture, an approach that remains underexplored in the urban public service literature, and shows that right-censored open complaints carry informative signals that only survival-based models can properly exploit. Practically, the model produces individual hazard curves that city governments can use to prioritise complaints at high risk of prolonged delay, and spatial hazard maps that support more targeted allocation of field resources before backlogs worsen. Because the required variable structure is identical, the framework is also directly replicable to other city complaint systems such as Jakarta CRM or Surabaya WargaKu once those datasets become publicly available.

Chapter 2 Research Methodology

A. Dataset Description

This study uses the Chicago 311 Service Requests dataset obtained from the official City of [Chicago Open Data Portal](#). The dataset contains public infrastructure complaint records submitted by residents between 2022 and 2026 and is widely used to analyze urban public service performance in the City of Chicago. The original dataset consists of more than 13.9 million records covering various categories of public complaints across the city. In this research, only four infrastructure-related complaint categories were selected as the study subset.

Table 1. Selected Infrastructure complaint categories

Category (sr_type)	Number of Records
Pothole in Street Complaint	317,589

Street Light Out Complaint	222,722
Tree Debris Clean-Up Request	60,441
Traffic Signal Out Complaint	49,342

These categories were selected because they represent infrastructure issues frequently reported by residents and contain clear spatial and temporal information relevant to this study. After the filtering process, the resulting dataset consists of 647,521 complaint records used for further analysis. The dataset also provides several important attributes, including complaint category, complaint submission time, completion time, complaint status, community area identifier, and geographic coordinates. In particular, the variables `created_date` and `closed_date` provide information regarding the submission and completion time of each complaint record, while `community_area`, `latitude`, and `longitude` provide spatial information related to the location of the reported infrastructure issue. In addition, the dataset also includes complaint status information, such as completed and unresolved reports, which reflects the condition of complaint handling within the public service system.

The Chicago 311 dataset was selected because it provides event-level complaint records with detailed temporal and spatial information that includes complaint submission time, completion time, and community area identifiers. These attributes make the dataset suitable for analyzing infrastructure complaint patterns across different regions and time periods.

B. Research Variable

Table 2. Research Variables

Variable Name	Description
<code>Sr_number</code>	Unique identifier for each complaint record.
<code>Sr_type</code>	Type of infrastructure complaint submitted by residents, used as a categorical feature in the model.
<code>Created_date</code>	Timestamp when the complaint was submitted, used as the start time in survival analysis
<code>Closed_date</code>	Timestamp when the complaint was completed. Absent complaints are still Open at the time of data collection.
<code>status</code>	Complaint handling status: Completed, Open, or Canceled. Used as the basis for constructing the event indicator.
<code>community_area</code>	Community area identifier in Chicago (1–77). Serves as the node ID in the spatial graph component.
<code>ward</code>	Political district identifier in Chicago (1–50), used as an additional feature.
<code>latitude</code>	A geographic coordinate used to verify node assignment in the spatial graph.
<code>longitude</code>	A geographic coordinate is used to verify node assignment in the spatial graph.
<code>duration_days</code>	Primary response variable representing the number of days from complaint submission to completion or censoring date for Open complaints.

event	Censoring indicator where 1 = Completed (observed event) and 0 = Open (right-censored), meaning the complaint had not yet been resolved at the time of data collection.
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C. Data Preprocessing

Before the dataset could be used for modelling, several preprocessing steps were carried out to ensure consistency, correctness, and compatibility with the survival analysis framework. The preprocessing pipeline was implemented in Python using the pandas library and ran directly against the raw CSV file retrieved from the Chicago Open Data Portal.

The first step addressed missing values. The approach differed by column, because not all missing entries carry the same meaning. Records with a missing community_area, latitude, or longitude were dropped outright. Since community_area serves as the node identifier in the spatial graph component of the model, any record without it cannot be assigned to a graph node and therefore holds no usable spatial information. The same applies to coordinates, which are used to verify node assignment. After removing these rows, the dataset shrank from 647,521 to 644,554 records, a reduction of roughly 0.5% from missing community areas and 0.2% from missing coordinates. Rows with a missing closed_date were handled differently: rather than being removed, they were retained. A missing closed_date simply indicates that the complaint was still open at the moment the data was extracted, not that the record is corrupt. Removing these rows would introduce survival bias by eliminating exactly the observations that survival analysis is designed to accommodate.

The second step computed the primary response variable, duration_days. Both created_date and closed_date were first

converted to datetime objects to enable arithmetic operations. For completed complaints, duration_days was calculated as the integer difference in calendar days between closed_date and created_date. For open complaints with no closed_date, the censoring date was defined as the maximum created_date observed in the dataset, which represents the effective extraction cutoff. The duration was then calculated as the difference between this censoring date and created_date. This follows the standard right-censoring convention in survival analysis: the complaint is known to have survived at least until the censoring date, even though its actual resolution time remains unobserved.

A small number of records produced a negative duration_days value after this calculation, indicating that their closed_date preceded their created_date. These entries reflect data entry errors rather than genuine observations and were removed before further analysis. No negative durations are plausible in the context of a public complaint system.

The binary event indicator was then assigned. A value of 1 was assigned to records with a status of Completed, indicating an observed resolution event. A value of 0 was assigned to all remaining records, including those with an Open status, treating them as right-censored observations consistent with the Cox model's assumptions. Canceled complaints were excluded from the analysis entirely, as they do not represent either a genuine resolution or a censored survival, and including them would distort the event-time distribution.

D. Dataset Exploration

An exploratory analysis was conducted on 647,521 rows of data after filtering for infrastructure categories and the 2022–2026 period. Overall data quality is high, reflecting a well-maintained recording system.

Table 3. Missing Values Summary

Column	Missing	Percentage
closed_date	9271	1.4261%
community_area	2950	0.4538%
ward	2792	0.4295%
latitude	1062	0.1634%
longitude	1062	0.1634%
parent_sr_number	544249	83.7185%
electricity_grid	3569	0.549%
electricity_district	413376	63.5871%

Missing values are minimal across all key columns. Rows with missing community_area will be dropped, while rows with missing closed_date correspond to Open complaints and are retained as right-censored observations.

Table 4. Descriptive Statistics

Statistics	Value
Valid observations	644,554
Mean duration	21.91 days
Median duration	4 days
75th percentile	15 days
Maximum	1,593 days
Censored (Open)	19,547 (3%)
Completed	625,007 (97%)

After dropping missing spatial entries and computing duration_days, the distribution is heavily right-skewed. Most complaints are resolved quickly, with 75% closed within two weeks, yet a long tail stretches well beyond four years, with a maximum of 1,593 days. The mean of 22 days sits far above the median of 4 days, a gap that

reflects a relatively small number of complaints accumulating disproportionately long resolution times. This kind of spread is precisely what makes the dataset well-suited for survival analysis, giving the model enough variation to learn what separates complaints that close quickly from those that linger.

Table 5. Median and Average Duration by Category

Category	Median Duration	Average Duration
Pothole in Street Complaint	7 Days	27 Days
Traffic Signal Out of Order Complaint	0 Days	17 Days
Tree Debris Clean-Up Request	4 Days	25 Days
Street Light Out Complaint	2 Days	12 Days

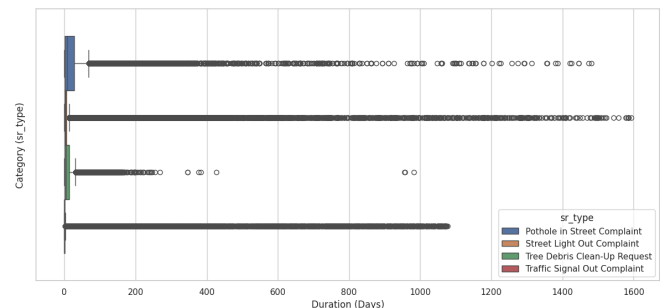


Figure 1. Boxplot of Resolution Duration

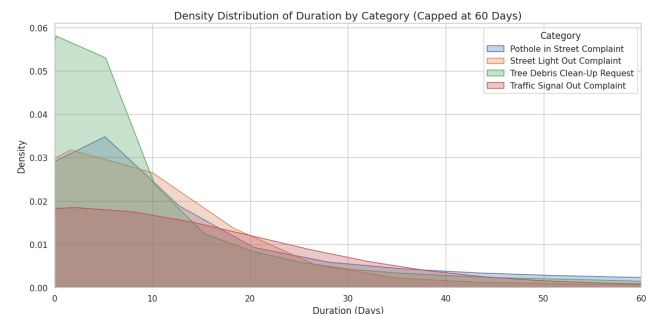


Figure 2. Distribution of Duration

Traffic Signal Out complaints are usually addressed in a day or two. Street Light Out cases are a different situation entirely, with a median that sits much further right and outliers stretching past 1,500 days. The

density chart, capped at 60 days, fills in what the boxplot compresses. Tree Debris Clean-Up requests spike near zero and are mostly gone by day ten. Street Light Out complaints barely peak at all; the distribution is flat from the start and bleeds into a long tail past 30 days. The most plausible read on that gap is parts. Signal repairs tend to need labor more than hardware. Street lights often need components the city doesn't stock, and waiting on suppliers shows up exactly where you'd expect it to.

E. Graph Convolutional Network (Spatial Components)

A Graph Convolutional Network is a deep learning method that operates directly on graph structures that allow each node to iteratively aggregate information from its neighbors. Different from conventional neural networks that work on structured data such as grids or text sequences, GCNs are designed to capture relational connections between arbitrarily linked entities, making them highly relevant for urban contexts where conditions in one area are influenced by pressures in surrounding areas.

In STG-SurviNet, the GCN represents Chicago's 77 community areas as nodes in a weighted graph $G = (V, E)$ where an edge is formed if two areas are geographically adjacent. Each node v is initialized with a feature vector x_v that includes the proportion of each complaint category, the average historical resolution time, and the density of complaints per square kilometer. A two-layer GCN then aggregates signals from neighbors so that overload in areas such as Englewood automatically propagates to the embeddings of areas directly bordering it, such as Gage Park and Chicago Lawn. The output of this module is a spatial embedding h_v of dimension d_s per node.

F. Temporal Convolutional Network (Temporal Component)

A Temporal Convolutional Network is an architecture that processes sequential data using dilated causal convolution, allowing the receptive field to grow exponentially without proportionally increasing the network's depth. Compared to LSTMs, TCNs are superior for long time windows because their training can be fully parallelized and they are not prone to the vanishing gradient problem.

For each node v , a time series $S_v = [c_{t-59}, c_{t-58}, \dots, c_t]$ is constructed, where c_t is the number of new complaints received on day t . A 60-day window is selected to capture a full operational cycle of the city department while remaining sensitive to seasonal changes, such as post-winter spikes in road damage. The output of this module is a d_t -dimensional temporal embedding h_t that represents the historical load conditions of the area before the arrival of new complaints.

G. Deep Cox Proportional Hazards (Survival Components)

The Cox Proportional Hazards model captures the relationship between covariates and the hazard function of an event, the instantaneous rate at which an event occurs at time t , given that the event has not yet occurred. Its main advantage is its ability to naturally handle right-censored observations, so that data for which the event has not yet occurred at the end of the observation period still contribute to the estimation. In the Deep Cox implementation, a neural network layer is used to learn non-linear representations of the covariates before passing them to the hazard function.

The two embeddings are combined via $z_v = \text{concat}([h_v, h_t])$ and then passed to a fully-connected network that produces the log-partial-hazard per observation. The loss function being minimized is the Cox Negative

Partial Log-Likelihood, given by

$$L(\theta) = - \sum_i [\varphi(z_i) - \log(\sum_{j: T_j \geq T_i} \exp(\varphi(z_j)))],$$

which ensures the 19,547 open cases continue to contribute to the hazard estimate. The final output is the individual survival curve $S(t|z_v)$, which shows the probability that a complaint remains unresolved on day t , conditional on the spatial and temporal profile of its origin.

H. Knowledge Discovery Framework and Evaluation

STG-SurviNet is treated as a feature learning tool from which three categories of patterns are extracted after training. The first is spatial patterns: GCN embeddings are clustered using GMM (Gaussian Mixture Model) to identify groups of neighborhoods with similar infrastructure stress profiles, independent of geographic proximity. The question being asked is whether administrative ward boundaries actually reflect service pressure or whether a different spatial structure emerges from the complaint data itself. The second is temporal patterns: the TCN's learned representations are analyzed to identify which time windows correlate most strongly with resolution delay, with particular attention to post-winter months when road damage complaints spike. The third is survival risk patterns: Cox hazard scores are aggregated per neighborhood and per complaint category to rank Chicago's 77 community areas by their structural resolution risk, separating neighborhoods where delays are driven by complaint volume from those where delays persist regardless of volume.

Table 6. Evaluation Metrics and Performance Targets

Metric	Description	Target
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C-index	Relative risk ranking across observations	≥ 0.75
Integrated Crier Score	Probability accuracy of the survival curve	< 0.20
D-Calibration	Calibration of predicted resolution time distribution	$p > 0.05$
MAE (completed only)	Prediction error in plain days	< 7 days

Model performance is validated against three baselines to confirm the embeddings are reliable enough to trust for pattern extraction: Kaplan-Meier as a covariate-free floor, Classic Cox PH using tabular features only, and Random Survival Forest as a non-linear tree-based benchmark. Three survival metrics are used: C-index (target ≥ 0.75), Integrated Brier Score (target < 0.20), and D-Calibration (target $p > 0.05$). A fourth metric, Mean Absolute Error on completed complaints, translates survival curve accuracy into plain days, which is more operationally readable than the probabilistic metrics alone. To confirm that the GCN and TCN each contribute meaningfully before pattern extraction begins, two reduced variants are tested as a brief ablation: TCN+Cox without the spatial component and GCN+Cox without the temporal component.

Table 7. Baseline Models

Model	Description
Kaplan-Meier	Non-parametric floor, no covariates
Classic Cox PH	Tabular-only parametric benchmark
Random Survival Forest	Non-linear tree-based benchmark

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Appendix

The files related to this analysis are available at the following Google Drive link:

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The dashboard and model testing could be accessed at the following link: [Dashboard STG-SurviNet](#)